

Graph Algorithms:

What is graph?

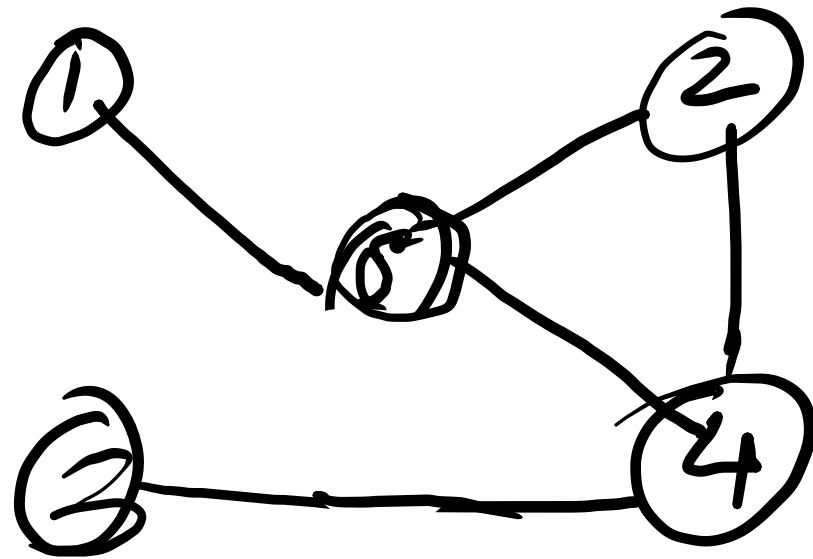
Set of vertices (V)

Set of edges (E)

an edge is represented as
 (u, v) where $u, v \in V$

Types of graphs

Undirected

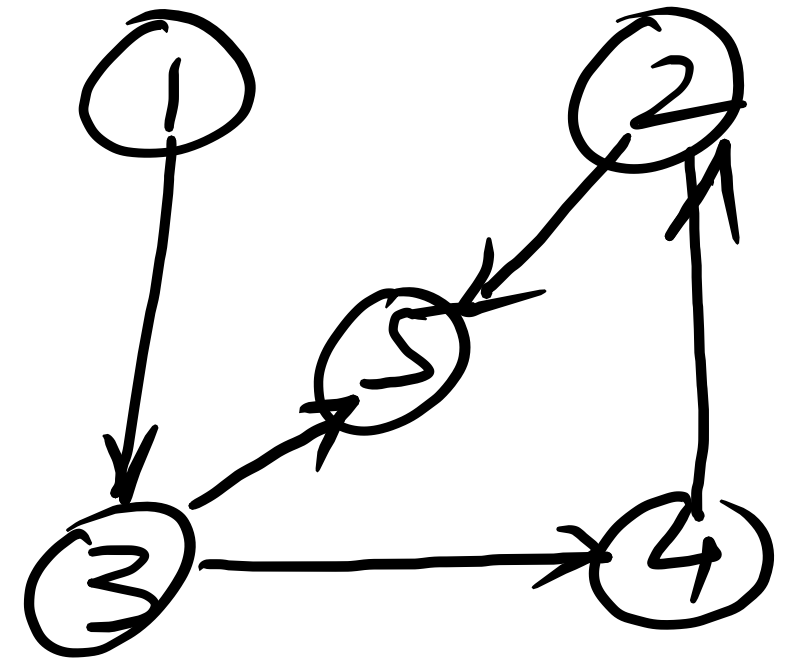


degree

$$d(v) = 3$$

$$d(1) = 1$$

Directed



in degree

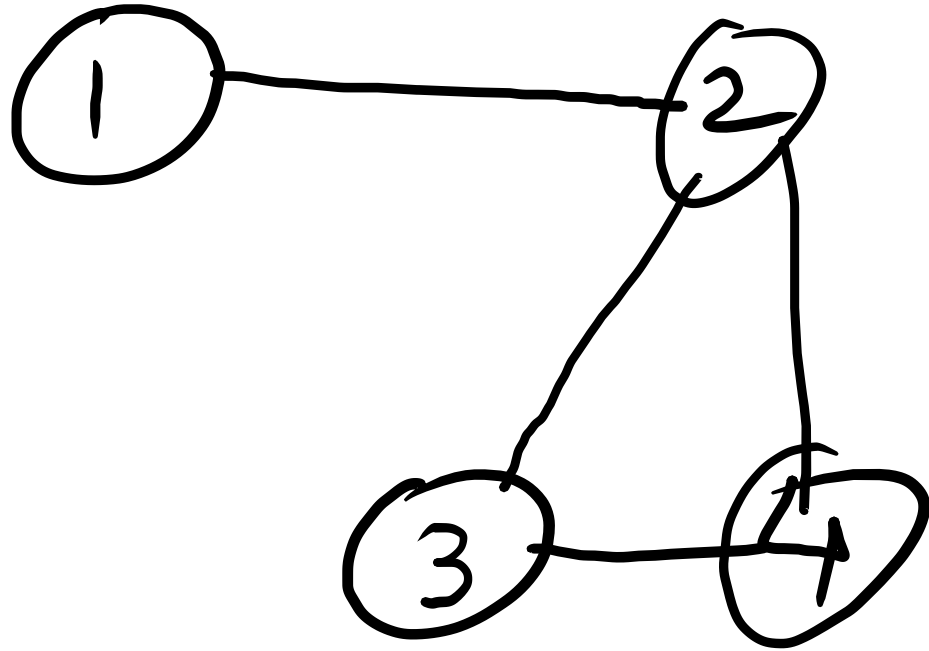
out degree

$$id(5) = 2$$

$$od(5) = 0$$

Graph Representations:-

Adjacency Matrix



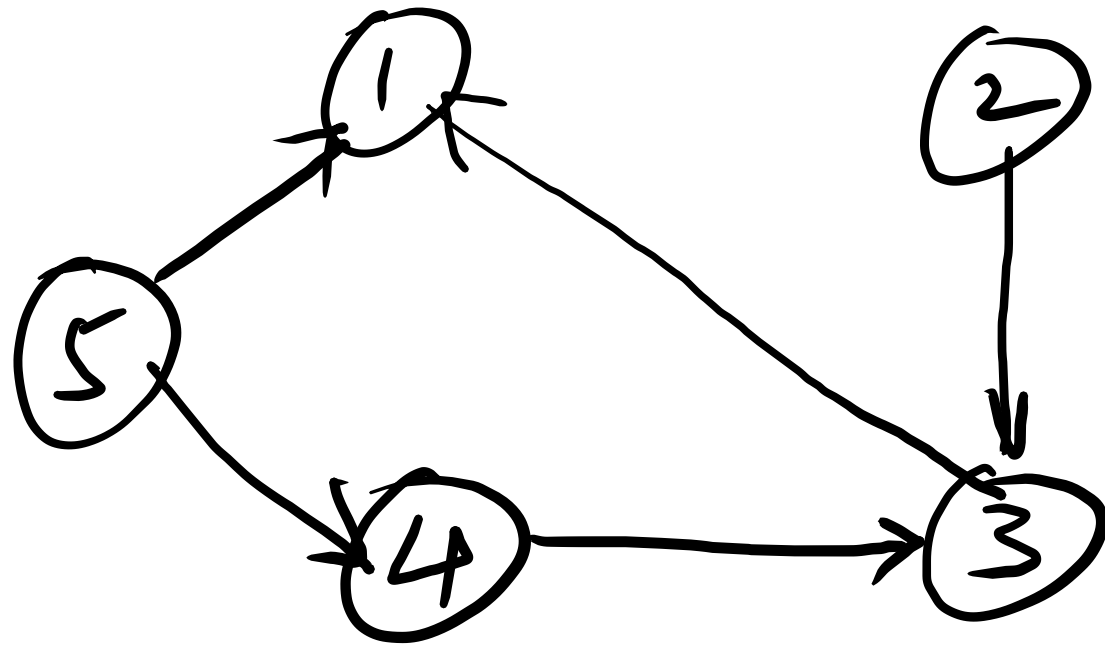
Undirected

	1	2	3	4
1		1		
2	1		1	1
3		1		1
4		1	1	

$$a_{ij} = \begin{cases} 1 & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$$

$i, j \in V$

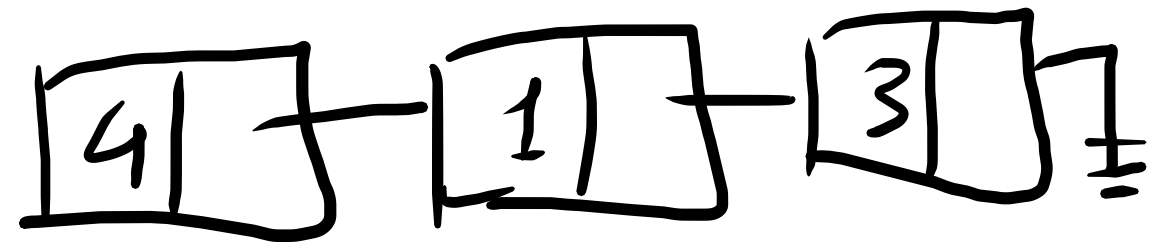
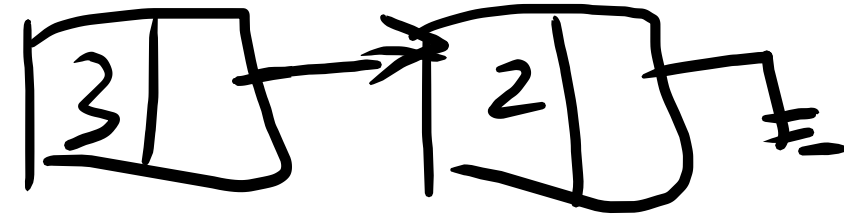
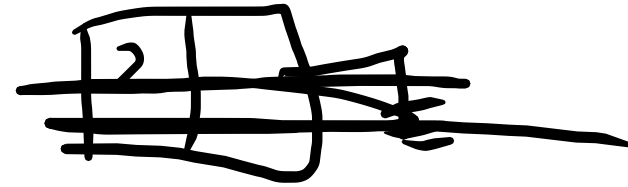
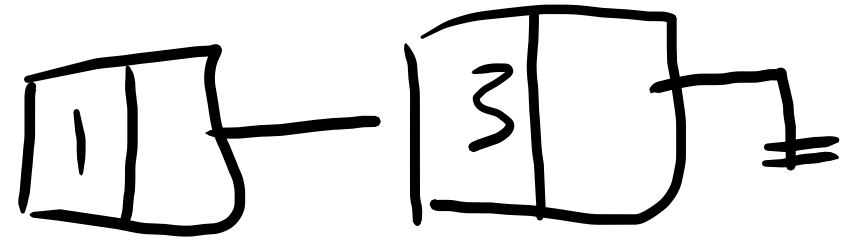
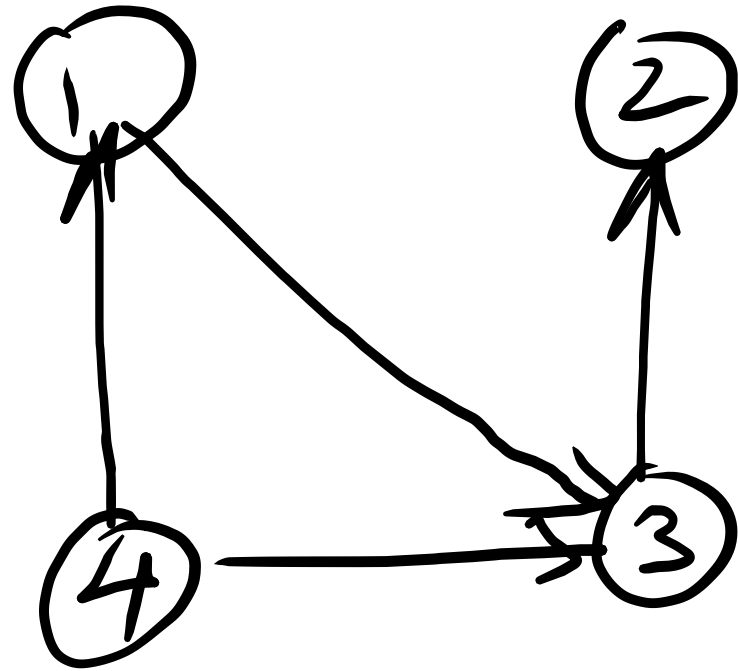
Directed



in \ out	1	2	3	4	5
1			1		1
2					
3		1		1	
4					1
5					

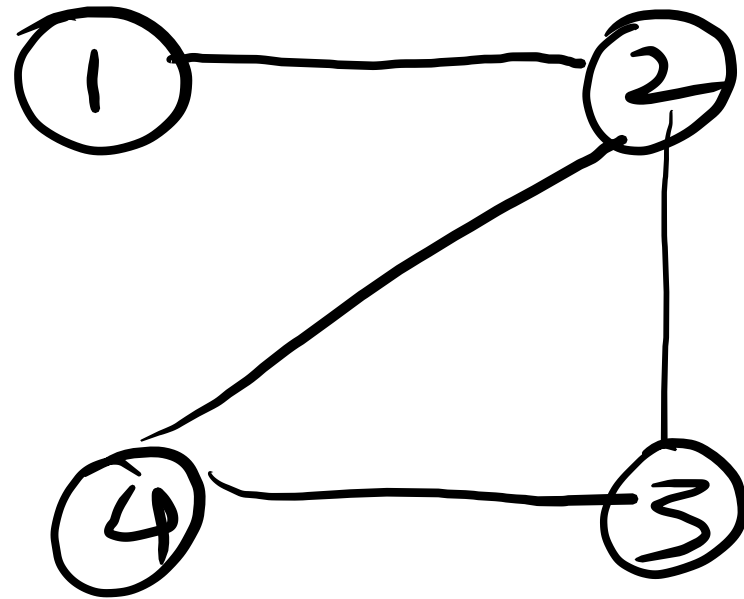
Space Complexity for Adjacency Matrix = $|V|^2$

Adjacency List : directed graph

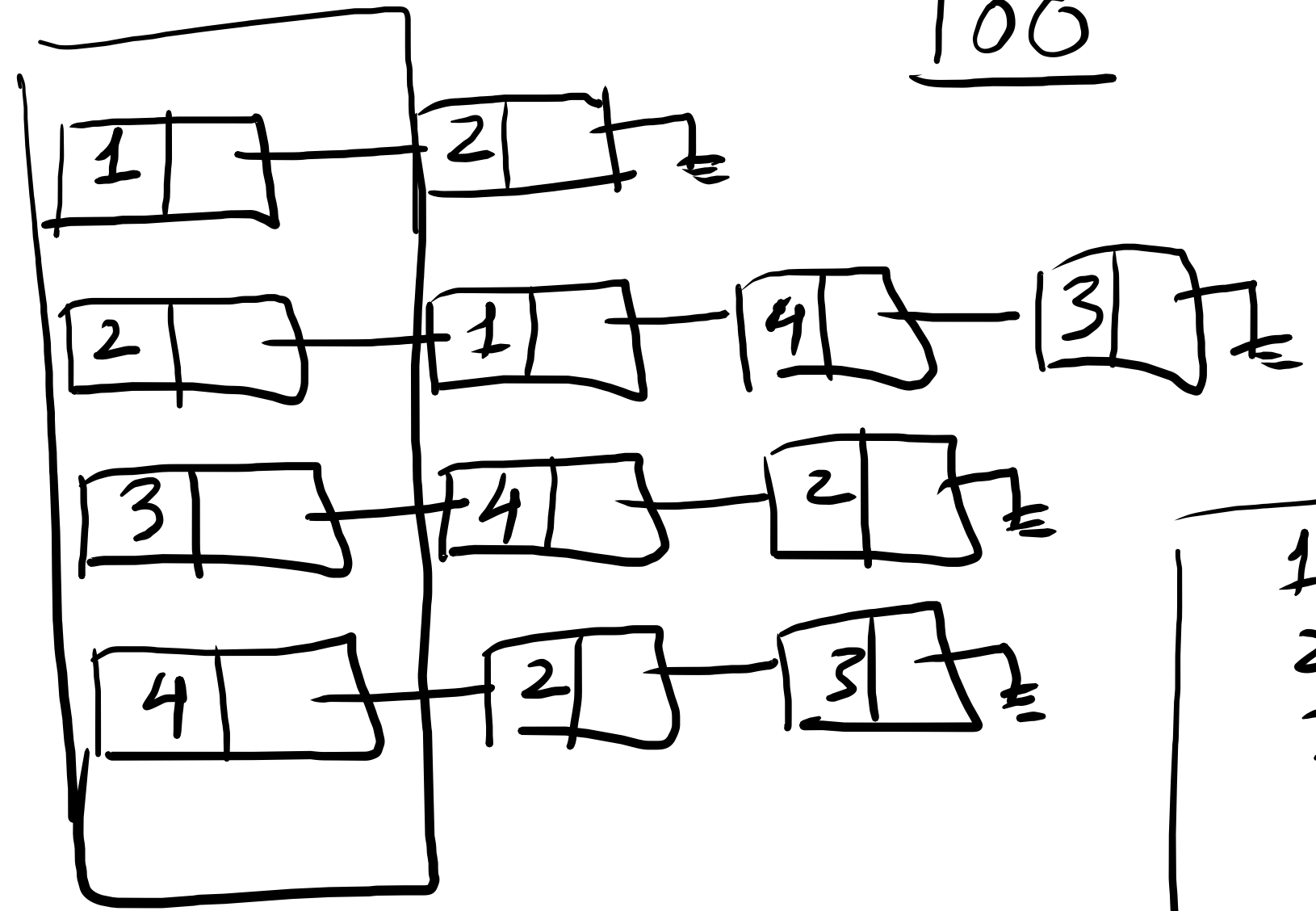


Space complexity = $O(|E|)$

Adjacency List



undirected



100

Space Complexity

$$2|E|$$

$$(u, v) \in E$$

